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The equation $x^4 + xy + 4y^3 - 2 = 0$ has no integer solutions

Size $h = 54$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + xy + 4y^3 - 2 = 0 \tag{1}$$

has no integer solutions.

The proof uses the quartic Hilbert reciprocity law over the field $K = \mathbb{Q}(i)$. To a hypothetical integer solution we attach the values of 43 auxiliary polynomials $H_0, \dots, H_{42}$ with coefficients in $\mathbb{Z}[i]$, constants $c_0, \dots, c_{42} \in \mathbb{Z}[i]$, and a sum B of quartic Hilbert symbols of these values, recorded by a graph with 164 weighted edges. By the reciprocity law, the local contributions B_p of all primes p add up to 0 modulo 4. We show that $B_p = 0$ for $p \notin S$, compute B_p for the primes $p \in S$, and obtain a contradiction. Two standard facts from class field theory are quoted without proof: the explicit formula for the quartic Hilbert symbol at the places not above 2, and the Hilbert reciprocity law. Apart from these, the proof uses polynomial identities, which can be checked by expanding both sides, and finite computations with residues.

1 Quartic Hilbert symbols over $\mathbb{Q}(i)$

The field. Let $K = \mathbb{Q}(i)$, where $i^2 = -1$. Its ring of integers is $\mathbb{Z}[i]$, and every element of K can be written uniquely as $A + Bi$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + Bi,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$. The units of $\mathbb{Z}[i]$ are ± 1 and $\pm i$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[i]$. A prime $p \equiv 3 \pmod{4}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[i]/p\mathbb{Z}[i]$ with p^2 elements. If $p \equiv 1 \pmod{4}$, the polynomial $t^2 + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which i is the root of $t^2 + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and i reduces to r . The prime 2 is ramified: $\pi = 1 + i$ generates the unique prime ideal above 2, $\pi^2 = 2i$, and $\mathbb{Z}[i]/\pi\mathbb{Z}[i] = \mathbb{F}_2$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the quartic Hilbert symbol $(a, b)_v$ is a fourth root of unity, see [1, Chapter V, §3]. We write $(a, b)_v = i^{\beta_v(a, b)}$ with $\beta_v(a, b) \in \mathbb{Z}/4\mathbb{Z}$, and we use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1) $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$, $\beta_v(a, b) = -\beta_v(b, a)$, and $\beta_v(a, c^4) = 0$. In particular, $\beta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 4}$.

(H2) Let v be a finite place not above 2, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/4\mathbb{Z}$ by $\bar{u}^{(q-1)/4} = \bar{i}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\beta_v(a, b) = e\chi(w) - f\chi(u) + ef\chi(-1)$, and $\chi(-1) = (q-1)/2 \pmod{4}$. In particular, $\beta_v(a, b) = 0$ if a and b are both units.

(H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\beta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \beta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\beta_v(a, b) = 0$.

The value of $\chi(-1)$ follows from $-1 = \bar{i}^2$ and $(-1)^{(q-1)/4} = \bar{i}^{(q-1)/2}$. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values β_v change sign, and the proof below is the same.

Classes at the places not above 2. Let $p \neq 2$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $c_v(a) = (e \pmod{4}, \chi(u)) \in (\mathbb{Z}/4\mathbb{Z})^2$. By (H2) and (H1), c_v is additive, $c_v(a)$ determines the class of a modulo fourth powers, and $\beta_v(a, b) = e\chi(w) - f\chi(u) + \varepsilon_v ef$ for $c_v(a) = (e, \chi(u))$ and $c_v(b) = (f, \chi(w))$, where $\varepsilon_v = \chi(-1)$. We put $c_p(a) = c_v(a) \in (\mathbb{Z}/4\mathbb{Z})^2$ if $p \equiv 3 \pmod{4}$; then $q = p^2 \equiv 1 \pmod{8}$ and $\varepsilon_v = 0$. If $p \equiv 1 \pmod{4}$, we put $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in (\mathbb{Z}/4\mathbb{Z})^4$, where $r < r'$ are the roots of $t^2 + 1$ in $\{1, \dots, p-1\}$; here $q = p$, and $\varepsilon_v = 0$ if $p \equiv 1 \pmod{8}$ and $\varepsilon_v = 2$ if $p \equiv 5 \pmod{8}$. Accordingly, β_p is the sum of the forms β_v over the places v above p . For $p \equiv 3 \pmod{4}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/4} = (\bar{u}^{p-1})^{(p+1)/4} = 1$ and $\chi(u) = 0$.

The place above 2.

Lemma 2. *Let v be the place above 2. Every non-zero element of K_v is equal to $\pi^a i^b (1 + 2i)^{c5^d}$ times a fourth power, for unique $a, b, c, d \in \mathbb{Z}/4\mathbb{Z}$, and every unit congruent to 1 modulo π^7 is a fourth power. Put $c_2(A) = (a, b, c, d) \in (\mathbb{Z}/4\mathbb{Z})^4$. Then c_2 is additive, and $\beta_v(A, B) = c_2(A)^\top M_2 c_2(B)$, where*

$$M_2 = \begin{pmatrix} 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 3 & 2 & 0 \\ 3 & 2 & 0 & 0 \end{pmatrix} \pmod{4}.$$

Proof. For $\alpha \in \mathbb{Z}[i]$, the polynomial $g(s) = s - 3(1-i)s^2 - 8is^3 + 4(1+i)s^4 + \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[i]$. Since $(1 + \pi^3 s)^4 = 1 - \pi^7(s - 3(1-i)s^2 - 8is^3 + 4(1+i)s^4)$, as one checks using $\pi^5 = -4 - 4i$, $\pi^6 = -8i$ and $\pi^7 = 8 - 8i$, we obtain $(1 + \pi^3 s)^4 = 1 + \pi^7 \alpha$. Hence every unit congruent to 1 modulo π^7 is a fourth power in K_v , and the class of a unit modulo fourth powers depends only on its residue modulo π^7 . The ring $\mathbb{Z}[i]/\pi^7 \mathbb{Z}[i]$ has 2^7 elements, 64 of which are units, and a direct enumeration shows that the 64 products $i^b (1 + 2i)^{c5^d}$ with $b, c, d \in \{0, 1, 2, 3\}$ are pairwise distinct modulo π^7 . Hence they represent all units modulo π^7 , the group of these units has exponent 4, and the fourth power of every unit is congruent to 1 modulo π^7 . Therefore two units are in the same class modulo fourth powers if and only if they are congruent modulo π^7 , and the 64 products represent the 64 classes of units. Since π^4 is a fourth power, the exponent of π is the valuation modulo 4, and the first statement follows.

By (H1), it remains to compute β_v on the 16 ordered pairs of the elements $\pi, i, 1 + 2i$ and 5. Their norms are 2, 1, 5 and 25, hence, for each such pair, all the symbols at the places not above 2 and 5 vanish by (H2), and, by (H3), β_v at the place above 2 is minus the sum of the values at the two places above 5. These places correspond to $\bar{i} = 2$ and $\bar{i} = 3$ in $\mathbb{F}_5$; there 5 is a uniformizer, $q = 5$, $\chi(u)$ is defined by $\bar{u} = \bar{i}^{\chi(u)}$, and $\chi(-1) = 2$. The classes (e, χ) of the four

elements at these places are

	$\bar{i} = 2$	$\bar{i} = 3$
π	$(0, 3)$	$(0, 2)$
i	$(0, 1)$	$(0, 1)$
$1 + 2i$	$(1, 3)$	$(0, 3)$
5	$(1, 0)$	$(1, 0)$

where, for $1 + 2i$ at the place with $\bar{i} = 2$, the unit part is $(1 + 2i)/5 \equiv (1 + 2 \cdot 7)/5 = 3 \pmod{5}$, using $i \equiv 7 \pmod{25}$. Formula (H2) and the sum over these two places give M_2 . $\square$

We write β_2 for the form of Lemma 2. Since $2 = i^3\pi^2$, we have $c_2(2) = (2, 3, 0, 0)$; also $c_2(-1) = (0, 2, 0, 0)$ and $c_2(5) = (0, 0, 0, 1)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 3. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[i]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $c_p(z) \in b + D$ for the vector b and the subgroup D given as follows.*

- (i) $p \equiv 3 \pmod{4}$. If $\zeta \neq 0$, then $b = c_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $(\mathbb{Z}/4\mathbb{Z})^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{4}$. For each root r , let ρ be the root of $t^2 + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $c_{v_r}(z)$ are those of $c_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 2$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = c_2(\zeta)$, and $D = 0$ if $m \geq 7$, while $D = D_m$ if $m \leq 6$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$, and $(0, 0, 0, 1)$; D_2 is spanned by $(0, 2, 0, 0)$, $(0, 0, 1, 0)$, and $(0, 0, 0, 1)$; D_3 is spanned by $(0, 2, 1, 0)$, $(0, 0, 2, 0)$, and $(0, 0, 0, 1)$; D_4 is spanned by $(0, 0, 2, 0)$ and $(0, 0, 0, 1)$; D_5 is spanned by $(0, 0, 2, 0)$ and $(0, 0, 0, 2)$; D_6 is spanned by $(0, 0, 0, 2)$. If $\zeta = 0$, then $b = 0$ and $D = (\mathbb{Z}/4\mathbb{Z})^4$.
- (iv) $p = 2$, $z \in \mathbb{Z}$, so that $B = 0$. If $2^k \nmid A$, let $j = v_2(A)$ and $u = A/2^j$. Then $b = c_2(A)$ and $D = 0$ if $k - j \geq 4$, while $b = j c_2(2) + c_2(u)$ and $D = R_{k-j}$ if $k - j \leq 3$, where R_1 is spanned by $(0, 2, 0, 0)$ and $(0, 0, 0, 1)$; R_2 is spanned by $(0, 0, 0, 1)$; R_3 is spanned by $(0, 0, 0, 2)$. If $2^k \mid A$, then $b = 0$ and D is spanned by $c_2(2)$, $c_2(-1)$ and $c_2(5)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'i$ and $A + Bi$ with $i \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 2, the class of a unit depends only on its residue modulo π^7 ; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo π^7 shows that their classes form D_m . (iv) If $2^k \nmid A$, then $z = 2^j u'$ with a rational unit $u' \equiv u \pmod{2^{k-j}}$, and u'/u is a rational unit congruent to 1 modulo 2^{k-j} . A rational unit congruent to 1 modulo 16 is congruent to 1 modulo π^8 , hence a fourth power, and the odd residues modulo 16 are $\pm 5^s$; hence the classes of the rational units congruent to 1 modulo 2^{k-j} form R_{k-j} , and this group is 0 if $k - j \geq 4$. If $2^k \mid A$, then also the valuation j is unknown, and every non-zero rational number is $2^j(\pm 5^s)$ times a fourth power. $\square$

Lemma 4. *Let p be a prime, and consider vertices i with vectors b_i , subgroups D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2, 3\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$*

for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i$, $d' \in D_j$. Then

$$\sum_{(i,j)} w_{ij} \beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in generating sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and $\beta_p(u, v) = -\beta_p(v, u)$, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

2 The graph

Put $F(x, y) = x^4 + xy + 4y^3 - 2$, so that (1) is the equation $F(x, y) = 0$. The certificate consists of 43 polynomials H_i with coefficients in $\mathbb{Z}[i]$ and constants $c_i \in \mathbb{Z}[i]$, listed in Table 1, and of 164 edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2, 3\}$, listed below. For an edge (i, j) we put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij} \bmod 4$, and $\rho_{ij} = 0$ if i and j are not joined by an edge. The edges $(i, j; w_{ij})$ are (0, 13; 3), (0, 27; 3), (0, 28; 3), (0, 31; 2), (0, 35; 2), (0, 37; 3), (0, 42; 2), (1, 10; 2), (1, 15; 2), (1, 16; 2), (1, 21; 3), (1, 22; 1), (1, 31; 3), (1, 33; 3), (1, 39; 1), (2, 14; 3), (2, 20; 2), (2, 22; 1), (2, 27; 3), (2, 28; 1), (2, 30; 2), (2, 31; 1), (2, 32; 1), (2, 37; 1), (3, 5; 3), (3, 8; 1), (3, 9; 1), (3, 10; 2), (3, 12; 1), (3, 19; 1), (3, 23; 1), (3, 30; 2), (3, 33; 1), (3, 36; 2), (3, 37; 1), (3, 42; 1), (4, 5; 1), (4, 10; 3), (4, 12; 3), (4, 22; 2), (4, 24; 3), (4, 25; 3), (5, 12; 1), (5, 20; 3), (5, 30; 3), (5, 39; 1), (5, 41; 1), (6, 13; 1), (6, 16; 3), (6, 17; 1), (6, 23; 2), (6, 24; 3), (6, 26; 3), (7, 16; 1), (7, 20; 3), (7, 27; 3), (7, 31; 1), (7, 36; 3), (8, 10; 3), (8, 17; 1), (8, 21; 1), (8, 24; 1), (8, 28; 3), (8, 33; 3), (8, 37; 1), (8, 38; 2), (9, 19; 3), (9, 21; 3), (9, 22; 1), (9, 23; 1), (9, 25; 3), (9, 31; 3), (9, 33; 1), (9, 39; 3), (10, 24; 3), (10, 25; 1), (10, 31; 1), (10, 35; 2), (10, 39; 2), (11, 23; 3), (11, 30; 2), (11, 31; 3), (11, 42; 3), (12, 22; 2), (12, 24; 3), (12, 31; 1), (12, 34; 3), (12, 36; 3), (12, 42; 3), (13, 20; 1), (13, 23; 1), (13, 26; 1), (13, 33; 3), (13, 41; 3), (14, 18; 1), (14, 25; 3), (14, 30; 3), (14, 34; 2), (14, 40; 1), (14, 42; 3), (15, 26; 2), (15, 42; 2), (16, 17; 3), (16, 23; 1), (16, 27; 1), (16, 29; 3), (16, 42; 2), (17, 29; 2), (17, 33; 1), (17, 39; 1), (17, 42; 3), (18, 20; 2), (18, 24; 2), (18, 25; 1), (18, 34; 1), (18, 39; 2), (18, 40; 1), (19, 20; 1), (19, 26; 3), (19, 30; 3), (20, 21; 2), (20, 24; 1), (20, 25; 2), (20, 26; 2), (20, 31; 2), (20, 33; 1), (20, 37; 2), (20, 39; 3), (20, 40; 1), (20, 41; 3), (21, 23; 3), (21, 25; 2), (21, 26; 2), (21, 30; 1), (21, 31; 2), (21, 32; 1), (22, 33; 1), (22, 41; 3), (23, 33; 1), (23, 35; 2), (24, 39; 3), (24, 40; 3), (25, 34; 1), (25, 39; 3), (26, 34; 2), (27, 32; 1), (27, 33; 2), (27, 36; 2), (28, 37; 2), (28, 41; 3), (29, 33; 1), (29, 39; 1), (30, 36; 2), (30, 37; 1), (30, 40; 3), (30, 41; 2), (31, 35; 2), (31, 37; 1), (32, 41; 1), (34, 35; 2), (34, 37; 2), (34, 39; 2), (37, 38; 2), (38, 39; 2). Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i *linear* if $H_i = \varepsilon_i(\lambda_i \eta - s_i(\xi))$ for a unit ε_i of $\mathbb{Z}[i]$, an element $\lambda_i \in \mathbb{Z}[i]$ whose norm is a power of 2, and a polynomial s_i with coefficients in $\mathbb{Z}[i]$; the last column of Table 1 gives $\lambda_i \eta = s_i(\xi)$ for the linear vertices, where $\lambda_i = 1$ if no factor is shown. For a linear vertex i , we put $f_i(t) = \lambda_i^d F$ and $g_{ij}(t) = \lambda_i^{d_j} H_j$ evaluated at $\xi = t$, $\eta = s_i(t)/\lambda_i$, where d and d_j are the degrees of F and H_j in η ; these are polynomials in t with coefficients in $\mathbb{Z}[i]$. For every linear vertex i , Table 2 gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[i]$ such that

$$D_i^4 \lambda_i^{e_i} c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^4 = f_i Q_i \quad (2)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[i]$, where $\rho_{ij} \in \{0, 1, 2, 3\}$; the table also gives an integer $e_i \geq 0$ with $e_i + \sum_j \rho_{ij} d_j \equiv 0 \pmod{4}$, and $e_i = 0$ if $\lambda_i = 1$; Q_i is the quotient of the division of the left side by f_i , and the remainder is zero. For the edges (i, j) listed in Table 3, the table gives a linear endpoint $k \in \{i, j\}$ and the factorization of the norm of the resultant $R_{ij} = \text{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using $i^2 = -1$.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) , which lie in $\mathbb{Z}[i]$.

Step 1: the values are non-zero. The vertex H_{26} is linear, and $H_{26} = 0$ would give $f_{26}(\xi) = 0$ with $\xi \in \mathbb{Z}$, where $f_{26}(t) = \langle 4t^3 + t - 1, 0 \rangle$. An integer root of $4t^3 + t - 1$ divides its constant term -1 , and $f_{26}(c) \neq 0$ for $c \in \{-1, 1\}$; hence $H_{26} \neq 0$. For $i \in \{1, 2, 5, 6, 7, 8, 9, 13, 14, 15, 16, 17, 20, 21, 24, 25, 27, 28, 29, 39, 40, 42\}$, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $H_i \neq 0$. For $i \in \{0, 11, 18, 19, 23, 31, 32, 33, 36, 37, 38, 41\}$, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $H_i \neq 0$. For $i \in \{22, 34\}$, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $H_i \neq 0$. For $i \in \{30, 35\}$, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $H_i \neq 0$. For $i \in \{3\}$, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $H_i \neq 0$. For $i \in \{10\}$, there is no residue pair (a, b) modulo 8 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{8}$, as one checks for the 8^2 residue pairs; hence $H_i \neq 0$. For $i \in \{4, 12\}$, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $H_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $H_i \neq 0$. Also, all c_i are non-zero.

Step 2: reciprocity. For every place v of K , put

$$B_v = \sum_{(i,j)} w_{ij} \beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/4\mathbb{Z}, \quad (3)$$

the first sum over the edges, and, for a rational prime p , put $B_p = \sum_{v|p} B_v$. Applying (H3) to every pair (H_i, H_j) with an edge, multiplied by w_{ij} , and to every pair (H_i, c_i) , and adding, we find that $B_v = 0$ for all but finitely many v and $\sum_p B_p = 0$, as the complex place contributes 0. By (H1) and the definition of c_p , $B_p = \sum_{(i,j)} w_{ij} \beta_p(c_p(H_i), c_p(H_j)) + \sum_i \beta_p(c_p(H_i), c_p(c_i))$, where β_p is the form of Lemma 2 if $p = 2$.

Step 3: the primes outside S . Let $S = \{2, 3, 5, 13, 17\}$. It contains 2 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides H_i and H_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $H_k = \varepsilon_k(\lambda_k \eta - s_k(\xi))$ and λ_k is a unit at v , we have $\eta \equiv s_k(t)/\lambda_k$ modulo v , hence $f_k(t) \equiv \lambda_k^d F(x, y) = 0$ and $g_{kl}(t) \equiv \lambda_k^{d_l} H_l \equiv 0$ modulo v . The resultant can be written as $R_{ij} = Af_k + Cg_{kl}$ with polynomials A and C with coefficients in $\mathbb{Z}[i]$, so v divides R_{ij} , a contradiction. By (H2), a term of (3) in which all arguments are units vanishes. Now let v divide H_i . Collecting the terms of (3) which contain H_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\beta_v(H_i, \Pi_i)$ with $\Pi_i = c_i \prod_j H_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^4 \Pi_i \equiv W^4$ modulo v for some $W \in K$ integral at v . If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv \lambda_i^{d_j} H_j$ and $f_i(t) \equiv 0$ modulo v as above, and (2) gives $D_i^4 \lambda_i^{E_i} \Pi_i \equiv w_i(t)^4$ modulo v , where $E_i = e_i + \sum_j \rho_{ij} d_j$ is divisible by 4; this gives the claim with $W = w_i(t)/\lambda_i^{E_i/4}$. Hence the reduction of Π_i is a non-zero fourth power in k_v , $\chi(\Pi_i) = 0$, and $\beta_v(H_i, \Pi_i) = v(H_i)\chi(\Pi_i) = 0$ by (H2). Hence $B_v = 0$, and $B_p = 0$.

Step 4: the primes in S . Let $p \in S$. Here 3 is inert, the places above 5 are v_2 and v_3 , with $\varepsilon_v = 2$, the places above 13 are v_5 and v_8 , with $\varepsilon_v = 2$, and the places above 17 are v_4

and v_{13} , with $\varepsilon_v = 0$. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 3, applied with ζ the residue of $H_i(a, b)$, gives a vector b_i and a subgroup D_i with $c_p(H_i) \in b_i + D_i$; when H_i has rational coefficients, $H_i \in \mathbb{Z}$, and we use this in (i) and (iv). With $\gamma_i = c_p(c_i)$, if the hypotheses of Lemma 4 hold, then B_p is determined by a , b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes and the value of B_p in each; every integer solution lies in one of them. The result is $B_2 \in \{0, 2, 3\}$, $B_3 = 0$, $B_5 = 3$, $B_{13} = 0$, $B_{17} = 0$.

Step 5: contradiction. By Steps 2 and 3, $\sum_{p \in S} B_p = 0$. But by Step 4, $\sum_{p \in S} B_p \in \{1, 2, 3\}$ in $\mathbb{Z}/4\mathbb{Z}$. This contradiction proves the theorem. $\square$

References

- [1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The polynomials H_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$. Here $\langle A, B \rangle = A + Bi$.

i	H_i	c_i	chart
0	$\langle -y + 1, 0 \rangle$	$\langle -1, 0 \rangle$	$y = \langle 1, 0 \rangle$
1	$\langle -y, 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, 1 \rangle$
2	$\langle -y - 1, -1 \rangle$	$\langle -2, 2 \rangle$	$y = \langle -1, -1 \rangle$
3	$\langle -x^2 - y, 0 \rangle$	$\langle -1, 1 \rangle$	$y = \langle -x^2, 0 \rangle$
4	$\langle -x^2 - y - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 1, 0 \rangle$
5	$\langle -2y + 1, 0 \rangle$	$\langle 0, 2 \rangle$	$2y = \langle 1, 0 \rangle$
6	$\langle -2y, -1 \rangle$	$\langle -2, 2 \rangle$	$2y = \langle 0, -1 \rangle$
7	$\langle -2y + 1, -1 \rangle$	$\langle -2, 2 \rangle$	$2y = \langle 1, -1 \rangle$
8	$\langle -2y - 1, 1 \rangle$	$\langle -1, -1 \rangle$	$2y = \langle -1, 1 \rangle$
9	$\langle -2y - 1, -1 \rangle$	$\langle 1, 1 \rangle$	$2y = \langle -1, -1 \rangle$
10	$\langle x - 2y, 0 \rangle$	$\langle -50, -25 \rangle$	$x = \langle 2y, 0 \rangle$
11	$\langle -x - 2y, 0 \rangle$	$\langle -2, 0 \rangle$	$x = \langle -2y, 0 \rangle$
12	$\langle -x - 2y + 1, 0 \rangle$	$\langle 1, 1 \rangle$	$x = \langle -2y + 1, 0 \rangle$
13	$\langle -x - 2y + 1, 1 \rangle$	$\langle 2, 2 \rangle$	$x = \langle -2y + 1, 1 \rangle$
14	$\langle -x - 2y - 1, 1 \rangle$	$\langle 0, 1 \rangle$	$x = \langle -2y - 1, 1 \rangle$
15	$\langle -x - 2y - 1, x \rangle$	$\langle 1, 0 \rangle$	$\langle 1, 1 \rangle x = \langle 0, -2y - 1 \rangle$
16	$\langle -x - 2y - 1, -x \rangle$	$\langle -3, -1 \rangle$	$\langle 1, 1 \rangle x = \langle -2y - 1, 0 \rangle$
17	$\langle -x - 2y - 1, -1 \rangle$	$\langle 0, -1 \rangle$	$x = \langle -2y - 1, -1 \rangle$
18	$\langle x - 2y + 1, x + 1 \rangle$	$\langle 2, 2 \rangle$	$\langle 1, 1 \rangle x = \langle 2y - 1, -1 \rangle$
19	$\langle -x - 2y + 1, -x + 1 \rangle$	$\langle -2, 0 \rangle$	$\langle 1, 1 \rangle x = \langle -2y + 1, 1 \rangle$
20	$\langle -2y + 1, -x^2 + x - 1 \rangle$	$\langle -2, 2 \rangle$	$2y = \langle 1, -x^2 + x - 1 \rangle$
21	$\langle -2y - 1, -x^2 - x + 1 \rangle$	$\langle 0, 1 \rangle$	$2y = \langle -1, -x^2 - x + 1 \rangle$
22	$\langle x, 0 \rangle$	$\langle -2, -2 \rangle$	$x = \langle 0, 0 \rangle$
23	$\langle x + 1, 0 \rangle$	$\langle 2, 2 \rangle$	$x = \langle -1, 0 \rangle$
24	$\langle x, 1 \rangle$	$\langle 1, -7 \rangle$	$x = \langle 0, -1 \rangle$
25	$\langle x, -1 \rangle$	$\langle 1, -3 \rangle$	$x = \langle 0, 1 \rangle$
26	$\langle x - 1, 0 \rangle$	$\langle 18, 0 \rangle$	$x = \langle 1, 0 \rangle$
27	$\langle x + 1, 1 \rangle$	$\langle 8, 6 \rangle$	$x = \langle -1, -1 \rangle$
28	$\langle x + 1, -1 \rangle$	$\langle -2, -11 \rangle$	$x = \langle -1, 1 \rangle$
29	$\langle x - 1, -1 \rangle$	$\langle -2, 2 \rangle$	$x = \langle 1, 1 \rangle$
30	$\langle x + 2, 0 \rangle$	$\langle 0, 3 \rangle$	$x = \langle -2, 0 \rangle$
31	$\langle x - 2, 0 \rangle$	$\langle 3, 4 \rangle$	$x = \langle 2, 0 \rangle$
32	$\langle x, -y \rangle$	$\langle 11, 2 \rangle$	$y = \langle 0, -x \rangle$
33	$\langle x, y \rangle$	$\langle 13, 9 \rangle$	$y = \langle 0, x \rangle$
34	$\langle x + y, 0 \rangle$	$\langle 0, 2 \rangle$	$y = \langle -x, 0 \rangle$
35	$\langle x - y + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, 0 \rangle$
36	$\langle x + y, -y \rangle$	$\langle 7, 1 \rangle$	$x = \langle -y, y \rangle$
37	$\langle x + y, y \rangle$	$\langle 18, 26 \rangle$	$x = \langle -y, -y \rangle$
38	$\langle x + 1, y + 1 \rangle$	$\langle -1, 0 \rangle$	$y = \langle -1, x + 1 \rangle$
39	$\langle x - y + 1, -y \rangle$	$\langle 1, 1 \rangle$	$x = \langle y - 1, y \rangle$
40	$\langle x + y + 1, y \rangle$	$\langle 1, 1 \rangle$	$x = \langle -y - 1, -y \rangle$
41	$\langle x + y, -y + 1 \rangle$	$\langle -11, -2 \rangle$	$x = \langle -y, y - 1 \rangle$
42	$\langle 2x + 1, 0 \rangle$	$\langle 15, 0 \rangle$	$2x = \langle -1, 0 \rangle$

Table 2: The data of the identities (2); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

$i = 0, D_i = 1, \rho_{ij}:$	13:3, 27:3, 28:3, 31:2, 35:2, 37:3, 42:2
w_i	$\langle -6t^3 + 9t + 10, 0 \rangle$
$i = 1, D_i = 1, \rho_{ij}:$	10:2, 15:2, 16:2, 21:3, 22:1, 31:3, 33:3, 39:1
w_i	$\langle -t^3 + 3t^2 + 6t - 2, -t^2 + 3t - 4 \rangle$
$i = 2, D_i = 1, \rho_{ij}:$	14:3, 20:2, 22:1, 27:3, 28:1, 30:2, 31:1, 32:1, 37:1
w_i	$\langle -3t^3 + 2t^2 + 12t + 22, -t^3 + 2t + 14 \rangle$

$i = 3, D_i = 2, \rho_{ij}: 5:3, 8:1, 9:1, 10:2, 12:1, 19:1, 23:1, 30:2, 33:1, 36:2, 37:1, 42:1$
$w_i \quad \langle -6t^5 + t^4 - t^3 + 6t^2 + 4t + 2, \\ -6t^5 + t^4 - t^3 + 6t^2 + 4t + 2 \rangle$
$i = 4, D_i = 1, \rho_{ij}: 5:1, 10:3, 12:3, 22:2, 24:3, 25:3$
$w_i \quad \langle -2t^4 - 3t^2, \\ 0 \rangle$
$i = 5, D_i = 12, e_i = 2, \rho_{ij}: 3:1, 4:3, 12:1, 20:3, 30:3, 39:1, 41:1$
$w_i \quad \langle 256t^3 + 64t^2 + 232t + 168, \\ 96t^3 - 96t^2 + 72t - 72 \rangle$
$i = 6, D_i = 1, e_i = 3, \rho_{ij}: 13:1, 16:3, 17:1, 23:2, 24:3, 26:3$
$w_i \quad \langle -8, \\ -4t + 4 \rangle$
$i = 7, D_i = 2, e_i = 1, \rho_{ij}: 16:1, 20:3, 27:3, 31:1, 36:3$
$w_i \quad \langle 8t^2 - 24t + 16, \\ -8t^2 + 32 \rangle$
$i = 8, D_i = 4, e_i = 2, \rho_{ij}: 3:3, 10:3, 17:1, 21:1, 24:1, 28:3, 33:3, 37:1, 38:2$
$w_i \quad \langle -192t^2 - 128t + 64, \\ 32t^2 + 96t \rangle$
$i = 9, D_i = 2, e_i = 3, \rho_{ij}: 3:3, 19:3, 21:3, 22:1, 23:1, 25:3, 31:3, 33:1, 39:3$
$w_i \quad \langle -160t^3 + 336t^2 + 176t - 224, \\ 320t^3 - 112t^2 - 304t - 160 \rangle$
$i = 10, D_i = 52, \rho_{ij}: 1:2, 3:2, 4:1, 8:1, 24:3, 25:1, 31:1, 35:2, 39:2$
$w_i \quad \langle 480t^3 + 620t^2 + 500t + 120, \\ -440t^3 - 200t^2 - 220t + 20 \rangle$
$i = 11, D_i = 1, \rho_{ij}: 23:3, 30:2, 31:3, 42:3$
$w_i \quad \langle -16t^2 + 4t + 4, \\ 0 \rangle$
$i = 12, D_i = 4, \rho_{ij}: 3:3, 4:1, 5:3, 22:2, 24:3, 31:1, 34:3, 36:3, 42:3$
$w_i \quad \langle 0, \\ -72t^3 + 116t^2 - 76t + 24 \rangle$
$i = 13, D_i = 4, \rho_{ij}: 0:1, 6:3, 20:1, 23:1, 26:1, 33:3, 41:3$
$w_i \quad \langle 8t^3 - 28t^2 + 8t - 4, \\ -24t^3 + 36t^2 - 12t + 12 \rangle$
$i = 14, D_i = 2, \rho_{ij}: 2:1, 18:1, 25:3, 30:3, 34:2, 40:1, 42:3$
$w_i \quad \langle -32t^3 - 36t^2 + 8, \\ 44t^2 + 36t + 8 \rangle$
$i = 15, D_i = 1, e_i = 8, \rho_{ij}: 1:2, 26:2, 42:2$
$w_i \quad \langle 12t^2 + 4t + 7, \\ 16t^3 + 12t^2 + 16t + 3 \rangle$
$i = 16, D_i = 2, e_i = 2, \rho_{ij}: 1:2, 6:1, 7:3, 17:3, 23:1, 27:1, 29:3, 42:2$
$w_i \quad \langle 56t^2 + 12t + 40, \\ -48t^3 - 32t^2 - 24t - 14 \rangle$
$i = 17, D_i = 1, \rho_{ij}: 6:3, 8:3, 16:1, 29:2, 33:1, 39:1, 42:3$
$w_i \quad \langle -32t^2 - 12t, \\ 16t^3 + 8t^2 - 16t - 4 \rangle$
$i = 18, D_i = 1, e_i = 2, \rho_{ij}: 14:3, 20:2, 24:2, 25:1, 34:1, 39:2, 40:1$
$w_i \quad \langle -48t^3 + 24t^2 + 8t, \\ 32t^3 + 40t^2 - 40t \rangle$
$i = 19, D_i = 2, e_i = 2, \rho_{ij}: 3:3, 9:1, 20:1, 26:3, 30:3$
$w_i \quad \langle 16t^3 + 16t^2 + 8t - 8, \\ -16t^3 - 16t^2 + 8t + 8 \rangle$
$i = 20, D_i = 1641354692, e_i = 0, \rho_{ij}: 2:2, 5:1, 7:1, 13:3, 18:2, 19:3, 21:2, 24:1, 25:2, 26:2, 31:2, 33:1, 37:2, 39:3, 40:1, 41:3$
$w_i \quad \langle 682803551872t^5 - 2993830958208t^4 + 9191586275200t^3 - 9559249726208t^2 + 5409905064832t \\ - 1785793904896, \\ -577756851584t^5 + 210093400576t^4 + 3624111159936t^3 - 8981492874624t^2 + 8456259373184t \\ - 9349156325632 \rangle$
$i = 21, D_i = 9826, e_i = 0, \rho_{ij}: 1:1, 8:3, 9:1, 20:2, 23:3, 25:2, 26:2, 30:1, 31:2, 32:1$
$w_i \quad \langle -27744t^5 - 152592t^4 - 23120t^3 + 184960t^2 + 161840t - 46240, \\ -85544t^5 + 60112t^4 + 289000t^3 + 124848t^2 - 175712t - 129472 \rangle$
$i = 22, D_i = 2, \rho_{ij}: 1:3, 2:3, 4:2, 9:3, 12:2, 33:1, 41:3$
$w_i \quad \langle 4t^2 - 2t + 6, \\ 8t^2 + 2t \rangle$
$i = 23, D_i = 1, \rho_{ij}: 3:3, 6:2, 9:3, 11:1, 13:3, 16:3, 21:1, 33:1, 35:2$
$w_i \quad \langle -4t^2 - 2t - 2, \\ -8t^2 - 4t \rangle$
$i = 24, D_i = 1, \rho_{ij}: 4:1, 6:1, 8:3, 10:1, 12:1, 18:2, 20:3, 39:3, 40:3$
$w_i \quad \langle 14t - 1, \\ 10t^2 + 7t - 8 \rangle$

$i = 25, D_i = 1, \rho_{ij}: 4:1, 9:1, 10:3, 14:1, 18:3, 20:2, 21:2, 34:1, 39:3$
$w_i \quad \langle 4t^2 - 3t - 7,$ $8t^2 - t + 1 \rangle$
$i = 26, D_i = 1, \rho_{ij}: 6:1, 13:3, 15:2, 19:1, 20:2, 21:2, 34:2$
$w_i \quad \langle 12t^2 + 6,$ $-6t^2 + 3t - 3 \rangle$
$i = 27, D_i = 2, \rho_{ij}: 0:1, 2:1, 7:1, 16:3, 32:1, 33:2, 36:2$
$w_i \quad \langle 4t^2 - 2t - 10,$ $8t^2 - 4t + 10 \rangle$
$i = 28, D_i = 2, \rho_{ij}: 0:1, 2:3, 8:1, 37:2, 41:3$
$w_i \quad \langle -6t^2 + 14,$ $2t^2 - 10t + 2 \rangle$
$i = 29, D_i = 1, \rho_{ij}: 16:1, 17:2, 33:1, 39:1$
$w_i \quad \langle 2t + 2,$ $2 \rangle$
$i = 30, D_i = 1, \rho_{ij}: 2:2, 3:2, 5:1, 11:2, 14:1, 19:1, 21:3, 36:2, 37:1, 40:3, 41:2$
$w_i \quad \langle 48t^2 - 132t + 108,$ $-24t^2 + 12t - 36 \rangle$
$i = 31, D_i = 1, \rho_{ij}: 0:2, 1:1, 2:3, 7:3, 9:1, 10:3, 11:1, 12:3, 20:2, 21:2, 35:2, 37:1$
$w_i \quad \langle -100t^2 + 50t + 750,$ $-200t^2 - 350t - 350 \rangle$
$i = 32, D_i = 1, \rho_{ij}: 2:3, 21:3, 27:3, 41:1$
$w_i \quad \langle 2t^3 - 3t^2 - 2t + 4,$ $-t^3 + 4t^2 - 4t - 2 \rangle$
$i = 33, D_i = 1, \rho_{ij}: 1:1, 3:3, 8:1, 9:3, 13:1, 17:3, 20:3, 22:3, 23:3, 27:2, 29:3$
$w_i \quad \langle 3112t^3 - 356t^2 - 252t - 4364,$ $-8654t^3 + 2242t^2 + 1104t - 1272 \rangle$
$i = 34, D_i = 1, \rho_{ij}: 12:1, 14:2, 18:3, 25:3, 26:2, 35:2, 37:2, 39:2$
$w_i \quad \langle -t^3 - 6t^2 - t + 2,$ $33t^3 + 12t^2 + 5t + 16 \rangle$
$i = 35, D_i = 1, \rho_{ij}: 0:2, 10:2, 23:2, 31:2, 34:2$
$w_i \quad \langle 0,$ $t^3 + 4t^2 + 4t \rangle$
$i = 36, D_i = 1, \rho_{ij}: 3:2, 7:1, 12:1, 27:2, 30:2$
$w_i \quad \langle 2t^2 - 2t,$ $4t^3 - 4t^2 + 2t - 2 \rangle$
$i = 37, D_i = 2, \rho_{ij}: 0:1, 2:3, 3:3, 8:3, 20:2, 28:2, 30:3, 31:3, 34:2, 38:2$
$w_i \quad \langle 54t^3 + 7t^2 - 42t - 36,$ $68t^3 - 91t^2 - 54t - 42 \rangle$
$i = 38, D_i = 1, \rho_{ij}: 8:2, 37:2, 39:2$
$w_i \quad \langle t^2 + t + 1,$ $1 \rangle$
$i = 39, D_i = 2, \rho_{ij}: 1:3, 5:3, 9:1, 10:2, 17:3, 18:2, 20:1, 24:1, 25:1, 29:3, 34:2, 38:2$
$w_i \quad \langle 40t^3 - 978t^2 + 577t + 43,$ $1048t^3 - 388t^2 - 311t - 85 \rangle$
$i = 40, D_i = 1, \rho_{ij}: 14:3, 18:3, 20:3, 24:1, 30:1$
$w_i \quad \langle 2t^2,$ $2t^2 - 2t + 2 \rangle$
$i = 41, D_i = 1, \rho_{ij}: 5:3, 13:1, 20:1, 22:1, 28:1, 30:2, 32:3$
$w_i \quad \langle 2t^3 - t^2 - 8t + 4,$ $-6t^3 + 13t^2 - 11t + 3 \rangle$
$i = 42, D_i = 32, e_i = 2, \rho_{ij}: 0:2, 3:3, 11:1, 12:1, 14:1, 15:2, 16:2, 17:1$
$w_i \quad \langle 1536t^2 + 192t - 528,$ $-1536t^2 - 192t + 528 \rangle$

Table 3: Edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
$(0, 13)$	0	$2 \cdot 5$	$(11, 42)$	11	2^{18}
$(0, 27)$	0	$2 \cdot 5$	$(12, 22)$	12	$2^6 \cdot 3^2$
$(0, 28)$	0	$2 \cdot 5$	$(12, 24)$	12	$2^7 \cdot 5$
$(0, 31)$	0	$2^4 \cdot 5^2$	$(12, 31)$	12	$2^6 \cdot 5^4$
$(0, 35)$	0	2^2	$(12, 34)$	12	2^2

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(0, 37)	0	$2 \cdot 5$	(12, 36)	12	$2^3 \cdot 5$
(0, 42)	0	5^4	(12, 42)	12	$2^{10} \cdot 5^2$
(1, 10)	1	$2^5 \cdot 5$	(13, 20)	13	$2^{16} \cdot 5 \cdot 17$
(1, 15)	1	5^3	(13, 23)	13	$2^9 \cdot 13$
(1, 16)	1	5	(13, 26)	13	2^8
(1, 21)	1	$2^3 \cdot 5^3$	(13, 33)	13	$2^6 \cdot 5 \cdot 13$
(1, 22)	1	$2^2 \cdot 5$	(13, 41)	13	$2^9 \cdot 17$
(1, 31)	1	$2^3 \cdot 5^2$	(14, 18)	14	$2^{10} \cdot 5 \cdot 13^2$
(1, 33)	1	$2 \cdot 5$	(14, 25)	14	$2^7 \cdot 5$
(1, 39)	1	$2^3 \cdot 5^3$	(14, 30)	14	$2^{12} \cdot 3^2$
(2, 14)	2	$2^5 \cdot 3^4 \cdot 5$	(14, 34)	14	$2^3 \cdot 13$
(2, 20)	2	$2^4 \cdot 5^2 \cdot 13 \cdot 17$	(14, 40)	14	2^5
(2, 22)	2	$2^2 \cdot 5^2$	(14, 42)	14	$2^{11} \cdot 3^2 \cdot 5$
(2, 27)	2	$2^3 \cdot 5$	(15, 26)	15	$2^{12} \cdot 3^2 \cdot 5$
(2, 28)	2	$2^4 \cdot 5$	(15, 42)	15	$2^{12} \cdot 3^6$
(2, 30)	2	$2^2 \cdot 3^2 \cdot 17$	(16, 17)	16	$2^{12} \cdot 3^2 \cdot 5$
(2, 31)	2	$2^2 \cdot 5^3$	(16, 23)	16	2^{13}
(2, 32)	2	2^6	(16, 27)	16	$2^{10} \cdot 13$
(2, 37)	2	$2^2 \cdot 13^2$	(16, 29)	16	2^{10}
(3, 5)	3	$2^4 \cdot 3^6$	(16, 42)	16	$2^{12} \cdot 3^6$
(3, 8)	3	$2^8 \cdot 5^2$	(17, 29)	17	$2^{11} \cdot 13$
(3, 9)	3	$2^8 \cdot 5^2$	(17, 33)	17	$2^6 \cdot 13$
(3, 10)	3	$2^8 \cdot 3^2 \cdot 5^2$	(17, 39)	17	$2^6 \cdot 5 \cdot 13^2$
(3, 12)	3	2^{16}	(17, 42)	17	$2^{11} \cdot 3^2 \cdot 5$
(3, 19)	3	$2^9 \cdot 5^2 \cdot 13$	(18, 20)	18	$2^{27} \cdot 5 \cdot 13^2$
(3, 23)	3	2^4	(18, 24)	18	$2^{13} \cdot 5$
(3, 30)	3	$2^2 \cdot 3^4 \cdot 13^2$	(18, 25)	18	$2^{14} \cdot 5$
(3, 33)	3	$2^3 \cdot 5$	(18, 34)	18	$2^{12} \cdot 13^2$
(3, 36)	3	$2^4 \cdot 5 \cdot 13$	(18, 39)	18	2^{16}
(3, 37)	3	$2^4 \cdot 5 \cdot 13$	(18, 40)	18	2^{16}
(3, 42)	3	$2^6 \cdot 3^2 \cdot 5^2$	(19, 20)	19	$2^{28} \cdot 13^2$
(4, 5)	4	$2^4 \cdot 3^2 \cdot 5^2$	(19, 26)	19	2^{12}
(4, 10)	4	$2^{10} \cdot 5^2$	(19, 30)	19	$2^{18} \cdot 13$
(4, 12)	4	$2^6 \cdot 3^6$	(20, 21)	20	2^{31}
(4, 22)	4	$2^2 \cdot 3^2$	(20, 24)	20	$2^7 \cdot 5$
(4, 24)	4	1	(20, 25)	20	2^6
(4, 25)	4	1	(20, 26)	20	$2^5 \cdot 3^2$
(5, 12)	5	$2^{12} \cdot 3^2$	(20, 31)	20	$2^9 \cdot 5$
(5, 20)	5	$2^{16} \cdot 13^2$	(20, 33)	20	$2^{17} \cdot 5^2$
(5, 30)	5	$2^4 \cdot 3^6$	(20, 37)	20	$2^{18} \cdot 13^2$
(5, 39)	5	$2^{10} \cdot 5 \cdot 13$	(20, 39)	20	$2^{24} \cdot 5 \cdot 13$
(5, 41)	5	$2^{10} \cdot 5 \cdot 13$	(20, 40)	20	$2^{21} \cdot 3^2$
(6, 13)	6	$2^{21} \cdot 5$	(20, 41)	20	$2^{19} \cdot 5^2 \cdot 13 \cdot 17$
(6, 16)	6	2^{17}	(21, 23)	21	2^5
(6, 17)	6	2^{15}	(21, 25)	21	$2^8 \cdot 5$
(6, 23)	6	2^7	(21, 26)	21	$2^5 \cdot 5$
(6, 24)	6	$2^5 \cdot 5$	(21, 30)	21	2^{14}
(6, 26)	6	2^6	(21, 31)	21	$2^9 \cdot 5^4$
(7, 16)	7	$2^{14} \cdot 3^2 \cdot 5 \cdot 13$	(21, 32)	21	2^{19}
(7, 20)	7	$2^{20} \cdot 3^2 \cdot 5$	(22, 33)	22	2^2
(7, 27)	7	$2^6 \cdot 5 \cdot 13$	(22, 41)	22	$2^4 \cdot 5$
(7, 31)	7	$2^9 \cdot 5^2$	(23, 33)	23	$2 \cdot 13$
(7, 36)	7	$2^{13} \cdot 5$	(23, 35)	23	1
(8, 10)	8	$2^{14} \cdot 5^2$	(24, 39)	24	$2^5 \cdot 5$
(8, 17)	8	$2^{16} \cdot 5 \cdot 13$	(24, 40)	24	2^7
(8, 21)	8	2^{20}	(25, 34)	25	2^4
(8, 24)	8	$2^5 \cdot 5$	(25, 39)	25	$2^4 \cdot 5$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(8, 28)	8	$2^6 \cdot 5^2$	(26, 34)	26	$2^2 \cdot 3^2$
(8, 33)	8	$2^{10} \cdot 5 \cdot 13$	(27, 32)	27	$2^4 \cdot 5$
(8, 37)	8	$2^{13} \cdot 5$	(27, 33)	27	$2^6 \cdot 5$
(8, 38)	8	$2^{10} \cdot 5^2$	(27, 36)	27	$2^4 \cdot 5^2$
(9, 19)	9	$2^{21} \cdot 5^2$	(28, 37)	28	$2^4 \cdot 5^2$
(9, 21)	9	$2^{27} \cdot 5$	(28, 41)	28	$2^3 \cdot 5^3$
(9, 22)	9	2^7	(29, 33)	29	2^6
(9, 23)	9	2^5	(29, 39)	29	$2^3 \cdot 13^2$
(9, 25)	9	$2^5 \cdot 5$	(30, 36)	30	$2^5 \cdot 13$
(9, 31)	9	$2^9 \cdot 5^2$	(30, 37)	30	$2^5 \cdot 13$
(9, 33)	9	$2^{10} \cdot 5^2$	(30, 40)	30	$2^7 \cdot 3^2$
(9, 39)	9	$2^{14} \cdot 5^2$	(30, 41)	30	$2^8 \cdot 17$
(10, 24)	10	$2^7 \cdot 5$	(31, 35)	31	2^{14}
(10, 25)	10	$2^7 \cdot 5$	(31, 37)	31	$2^5 \cdot 5^3$
(10, 31)	10	$2^{12} \cdot 5^2$	(32, 41)	32	$2 \cdot 5$
(10, 35)	10	$2^4 \cdot 3^2$	(34, 35)	34	3^6
(10, 39)	10	$2^4 \cdot 5^2$	(34, 37)	34	2^2
(11, 23)	11	2^8	(34, 39)	34	$2^4 \cdot 13^2$
(11, 30)	11	2^{16}	(37, 38)	37	$2^2 \cdot 5^2$
(11, 31)	11	2^{14}	(38, 39)	38	$2^6 \cdot 13^2$

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p .

p	k	B_p	classes (a, b)
2	5	3	(1, 13), (3, 7), (5, 9), (7, 3), (9, 5), (11, 31), (13, 1), (15, 27), (17, 29), (19, 23), (21, 25), (23, 19), (25, 21), (27, 15), (29, 17), (31, 11)
2	6	0	(2, 3), (2, 35), (10, 31), (10, 63), (18, 27), (18, 59), (26, 23), (26, 55), (34, 19), (34, 51), (42, 15), (42, 47), (50, 11), (50, 43), (58, 7), (58, 39)
2	6	2	(6, 21), (6, 53), (14, 17), (14, 49), (22, 13), (22, 45), (30, 9), (30, 41), (38, 5), (38, 37), (46, 1), (46, 33), (54, 29), (54, 61), (62, 25), (62, 57)
3	1	0	(0, 2), (1, 2)
5	2	3	(1, 13), (5, 22), (7, 11), (10, 2), (17, 16), (20, 12), (22, 6)
5	3	3	(0, 92), (2, 121), (6, 123), (11, 33), (12, 1), (15, 57), (16, 43), (21, 28), (25, 117), (27, 7), (27, 22), (27, 32), (27, 47), (27, 57), (27, 71), (27, 72), (27, 82), (27, 97), (27, 107), (27, 122), (31, 48), (36, 83), (37, 76), (40, 82), (41, 93), (46, 78), (50, 17), (52, 2), (52, 21), (52, 27), (52, 52), (52, 77), (52, 102), (61, 8), (62, 26), (65, 107), (66, 18), (71, 3), (75, 42), (77, 12), (77, 17), (77, 37), (77, 42), (77, 62), (77, 67), (77, 87), (77, 92), (77, 96), (77, 112), (77, 117), (81, 23), (86, 58), (87, 101), (90, 7), (91, 68), (96, 53), (100, 67), (102, 46), (106, 73), (111, 108), (112, 51), (115, 32), (116, 118), (121, 103)
5	4	3	(56, 473), (181, 98), (306, 348), (431, 598), (556, 223)
13	1	0	(1, 7), (6, 9), (8, 12), (10, 12), (12, 3)
13	2	0	(3, 119), (11, 126), (12, 5), (12, 18), (12, 31), (12, 44), (12, 57), (12, 70), (12, 83), (12, 96), (12, 109), (12, 122), (12, 135), (12, 148), (12, 161), (16, 28), (20, 37), (24, 100), (29, 106), (33, 128), (42, 15), (46, 50), (50, 48), (55, 93), (59, 141), (63, 22), (68, 2), (72, 63), (74, 43), (76, 165), (81, 80), (85, 154), (89, 139), (94, 158), (98, 76), (102, 113), (107, 67), (111, 167), (115, 87), (120, 145), (124, 89), (128, 61), (133, 54), (137, 11), (140, 7), (140, 33), (140, 46), (140, 59), (140, 72), (140, 85), (140, 111), (140, 124), (140, 137), (140, 150), (140, 163), (141, 35), (146, 132), (150, 102), (152, 108), (154, 9), (159, 41), (163, 24), (167, 152)

p	k	B_p	classes (a, b)
13	3	0	(7, 1974), (9, 17), (22, 1915), (35, 264), (37, 1426), (48, 1655), (61, 1694), (87, 2110), (100, 290), (113, 1512), (126, 1382), (139, 2097), (165, 1668), (176, 960), (178, 524), (191, 225), (204, 771), (206, 1088), (217, 2162), (230, 4), (256, 420), (269, 797), (282, 2019), (295, 1889), (308, 407), (334, 2175), (345, 2143), (347, 1031), (360, 732), (373, 1278), (375, 750), (386, 472), (399, 511), (425, 927), (438, 1304), (451, 329), (464, 199), (477, 914), (503, 485), (514, 1129), (516, 1538), (529, 1239), (542, 1785), (544, 412), (555, 979), (568, 1018), (594, 1434), (607, 1811), (620, 836), (633, 706), (646, 1421), (672, 992), (683, 115), (685, 2045), (698, 1746), (711, 95), (713, 74), (724, 1486), (737, 1525), (763, 1941), (776, 121), (789, 1343), (802, 1213), (815, 1928), (841, 1499), (852, 1298), (854, 355), (867, 56), (880, 602), (882, 1933), (893, 1993), (906, 2032), (932, 251), (945, 628), (958, 1850), (971, 1720), (984, 238), (1010, 2006), (1021, 284), (1023, 862), (1036, 563), (1049, 1109), (1051, 1595), (1062, 303), (1075, 342), (1101, 758), (1114, 1135), (1127, 160), (1140, 30), (1153, 745), (1179, 316), (1190, 1467), (1192, 1369), (1205, 1070), (1218, 1616), (1220, 1257), (1231, 810), (1244, 849), (1270, 1265), (1283, 1642), (1296, 667), (1309, 537), (1322, 1252), (1348, 823), (1359, 453), (1361, 1876), (1374, 1577), (1387, 2123), (1389, 919), (1400, 1317), (1413, 1356), (1439, 1772), (1452, 2149), (1465, 1174), (1478, 1044), (1491, 1759), (1517, 1330), (1528, 1636), (1530, 186), (1543, 2084), (1556, 433), (1558, 581), (1569, 1824), (1582, 1863), (1608, 82), (1621, 459), (1634, 1681), (1647, 1551), (1660, 69), (1686, 1837), (1697, 622), (1699, 693), (1712, 394), (1725, 940), (1727, 243), (1738, 134), (1751, 173), (1777, 589), (1790, 966), (1803, 2188), (1816, 2058), (1829, 576), (1855, 147), (1866, 1805), (1868, 1200), (1881, 901), (1894, 1447), (1896, 2102), (1907, 641), (1920, 680), (1946, 1096), (1959, 1473), (1972, 498), (1985, 368), (1998, 1083), (1999, 20), (1999, 98), (1999, 189), (1999, 267), (1999, 358), (1999, 436), (1999, 527), (1999, 605), (1999, 696), (1999, 774), (1999, 865), (1999, 943), (1999, 1034), (1999, 1112), (1999, 1203), (1999, 1281), (1999, 1372), (1999, 1450), (1999, 1541), (1999, 1619), (1999, 1710), (1999, 1788), (1999, 1879), (1999, 1957), (1999, 2048), (1999, 2126), (2024, 654), (2035, 791), (2037, 1707), (2050, 1408), (2063, 1954), (2065, 1764), (2076, 1148), (2089, 1187), (2115, 1603), (2128, 1980), (2141, 1005), (2154, 875), (2167, 1590), (2193, 1161)
17	1	0	(0, 15), (1, 9), (4, 3), (8, 15), (9, 4), (10, 14), (11, 1), (12, 10), (13, 7), (14, 4)
17	2	0	(15, 182), (32, 284), (49, 97), (66, 199), (83, 12), (100, 114), (117, 216), (134, 29), (151, 131), (168, 233), (185, 46), (202, 148), (219, 250), (236, 63), (253, 165), (270, 267), (287, 80)